Community-Based Early Warning System (EWS) for Water-Borne Diseases

WaterGuard Platform — Comprehensive Project Documentation

1. Abstract

Waterborne disease outbreaks remain a critical threat to public health in rural and peri-urban communities. Traditional disease tracking models are reactive, relying on clinical logs that only record cases after an epidemic has already spread. WaterGuard is a proactive, crowdsourced Early Warning System (EWS) designed to detect potential outbreaks before they become widespread. The system integrates crowdsourced symptom surveys, local water source monitoring, and machine learning models to identify environmental-clinical risk correlations. A spatial-temporal audit rule automatically flags clusters (3 or more cases in the same village/district within 7 days), generating warnings that are instantly dispatched to citizens and ASHA field workers. By bridging environmental parameters (pH, turbidity, dissolved oxygen) with clinical observations, WaterGuard answers the core question: Can we identify early signs of a waterborne outbreak and warn people before it becomes serious?

2. Introduction & Problem Statement

2.1 Introduction

Access to safe drinking water is a fundamental human need and right. However, biological contamination of water infrastructure frequently triggers sudden outbreaks of gastrointestinal diseases. In rural jurisdictions, environmental water safety and clinical healthcare reporting exist in separate databases. An Early Warning System (EWS) acts as a digital bridge between clinical observations and environmental hazards. By analyzing symptoms at the community level in real-time, the system detects micro-anomalies and generates localized warnings, enabling community containment before local clinics become overwhelmed.

2.2 Problem Statement

Existing water security and healthcare surveillance systems suffer from three primary flaws:

1. **Reporting Lag:** Clinical health channels only log cases after patients visit hospitals, leading to a reporting delay of 3 to 7 days, by which time waterborne outbreaks are already widespread.
2. **Siloed Data:** Physical water parameters (pH, turbidity, dissolved oxygen) and health indicators (gastrointestinal symptoms) are monitored independently, hiding critical correlations.
3. **Loss of Explainability:** Standard machine learning models often use lossy preprocessing techniques (like PCA) that mask physical thresholds, making predictions hard for health officials to interpret and act on.

3. Solution (Platform Overview)

The WaterGuard EWS coordinates four core stakeholder dashboards to streamline regional outbreak surveillance:

3.1 Public Landing Page: Explains the 4-step EWS workflow, presents a live-updating stream of automated cluster alerts, and provides targeted dashboard entry points based on user credentials.

3.2 Villager Dashboard (Citizen Reporting & Warning Center): Allows citizens to log symptoms and specify water sources, view their latest risk assessment with explainable AI drivers, and access active local warnings. Features a persistent list of district emergency contacts.

3.3 ASHA Worker Dashboard (Field Verification & Referral): Ground verification portal where workers conduct home visits, submit clinical diagnoses, manage Primary Health Centre (PHC) referrals, log visit logs, and maintain district contact lists.

3.4 Health Official Dashboard (District Analytics & Broadcasts): Analytics board showing Telangana district trends. Features active Leaflet map hotspots, machine learning driver metrics, manual SMS alert triggers, and an Outbreak Simulator sandbox.

4. Technical Implementation Workflow

The platform operates via a structured five-phase containment loop:

- **Phase 1 - Localized Input Logging:** Citizens submit symptom reports (diarrhea, vomiting, fever, abdominal pain) and specify water sources via the client app.
- **Phase 2 - Feature Assembly:** The backend maps the citizen location and fetches the latest water quality indices (pH, turbidity, nitrates) for environmental correlation.
- **Phase 3 - Machine Learning Assessment:** The merged vector is evaluated by an optimized Gradient Boosting classifier (without PCA) to compute outbreak risk (Low, Medium, High).
- **Phase 4 - Spatial Cluster Auditing:** A database trigger evaluates rolling reports. If 3 or more symptomatic reports occur in the same village/district in a 7-day window, a cluster outbreak threat is flagged.
- **Phase 5 - Emergency Warning Dispatch:** Critical alerts and safe-water recommendations are broadcasted to villagers via in-app feeds and SMS alert logs.
- **Phase 6 - Field Action & Ground Verification:** ASHA workers conduct ground visits, record clinical diagnoses, refer high-risk cases to PHCs, and update the ML training loop.

5. Limitations

1. **Connectivity Reliance:** The crowdsourced warning pipeline requires mobile network or internet access, which can be limited in remote rural regions.
2. **False Positives:** Crowdsourced reporting can result in false reports. This is mitigated by requiring ASHA field workers to verify cases on the ground before they are officially classified as confirmed outbreaks.
3. **Static Environmental Parameters:** Water quality tests are often uploaded periodically (monthly/weekly), meaning the ML model might correlate real-time symptoms with slightly outdated physical water statistics.
4. **Local Gateway Integration:** The SMS notification engine is currently simulated via mock logs and requires integration with regional cellular gateways (e.g. Twilio) for physical deployments.

6. Future Enhancements

1. **Real-time IoT Sensor Integration:** Deploy solar-powered IoT water sensors in village storage tanks to transmit continuous pH, turbidity, and chlorine metrics directly to the EWS backend.
2. **Advanced Spatial Clustering Algorithms:** Replace rule-based counts with spatial algorithms (like DBSCAN) to model outbreak clusters across administrative borders.
3. **Speech-to-Text Multi-Lingual Reporting:** Integrate IVR phone lines allowing rural citizens to report symptoms by speaking in their native dialect.
4. **National Database Integration:** Establish API sync pipelines with national clinical networks, such as India's Integrated Disease Surveillance Program (IDSP), to share telemetry and coordinate containment protocols automatically.